

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (IV)	Branch: Computer Science & Engineering
Subject: Design & Analysis of Algorithms	Course Code: B022415(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

UNIT I Algorithms: Definitions and Application of notations, Asymptotic notations: big oh, small oh, omega and theta notations, worst case, best case and average case analysis. solving recurrence equations: General recurrence equation, Master Method, Recursive Tree Method, substitution method, analysing control structures. Analysis of Sorting and Searching: Heap, insertion, selection and bubble sort; sequential, binary and Fibonacci search.

UNIT II Divide-Conquer & Greedy Paradigm: Introduction to Divide and conquer paradigm, Quick and merge sorting techniques, the basic divide and conquer algorithm for matrix multiplication, Greedy Method: The basic greedy strategy & computing minimum spanning trees, Algorithms of Kruskal and Prims, use of greedy strategy in algorithms for the Knapsack problem and Huffman trees.

UNIT III Dynamic Programming and String-Matching Algorithms: The basic dynamic programming paradigm, Dynamic programming solution to the optimal matrix chain multiplication and the longest common subsequence problems, String matching algorithm: The general string problem as a finite automata, Knuth Morris and Pratt algorithms, Boyer-Moore Algorithm, linear time analysis of KMP algorithm and Boyer-Moore algorithm.

UNIT IV Backtracking: Back tracking and Recursive back tracking, the general method, 8-queens problem, sum of subsets, graph coloring, Hamiltonian cycle, Knapsack problem

UNIT V Branch and Bound & NP Complete Problem: General method, applications: Travelling sales person problem, 0/1 knapsack problem, LC (Least-cost search), FIFO Branch and Bound solution. NP-Hard and NP-Complete problems: Basic concepts, non-deterministic algorithms, NP – Hard and NP-Complete classes, Cook's theorem.

Text Books:

1. Cormen, Lelerson, Rivert, "Introduction to Algorithms", Second Edition, PHI.
2. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, "Fundamentals of Computer Algorithms", Galgotia Publications Pvt. Ltd., 2008

Reference Books:

1. Paneerselvam, "Design and Analysis of Algorithms", Prentice-Hall of India, 2006
2. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", Pearson Education, 2005.
3. Gilles Brassard and Paul Bratley, "Fundamentals of Algorithms", Prentice-Hall of India, 1997

Course Outcomes: [The students should be able to]:

1. Calculate the Time complexity of Insertion sort, Heap sort, Bubble sort, Linear and Binary search algorithms.
2. Apply the algorithms and design techniques to solve problems related to divide and conquer and Greedy Algorithm
3. Analyse Dynamic programming problems including Matrix chain multiplication, Longest Common subsequence and Knapsack Problem.
4. Understand the implementation of Backtracking and Recursive Backtracking Methods.
5. Understand the basic concepts of NP-Hard, NP-Complete and Branch and Bound methods.



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**B. Tech. (Fourth Semester) Examination,
April-May 2023
(AICTE Scheme)**

(Computer Science Engg. Branch)

DESIGN & ANALYSIS of ALGORITHMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Marks : 35

***Note : Attempt all questions. Part (a) is compulsory
from each unit. Attempt any two part from
(b), (c) and (d).***

Unit-I

1. (a) Define different asymptomatic notations with examples. 4

(b) Using master method find the solution of the following recurrence equations : 8

(i) $T(n) = 9T(n/3) + n$

(ii) $T(n) = T(2n/3) + 1$

(c) Write an algorithm for selection sort and find its asymptomatic complexity. 8

(d) What is a recursion tree? Using recursion tree find the asymptomatic bound for the equation : 8

$$T(n) = 3T(\lfloor n/4 \rfloor) + \Theta(n^2)$$

Unit-II

2. (a) Explain the difference between Divide & Conquer paradigm and greedy method. 4

(b) Explain Strassen's matrix multiplication method and analyze the efficiency of the algorithm. 8

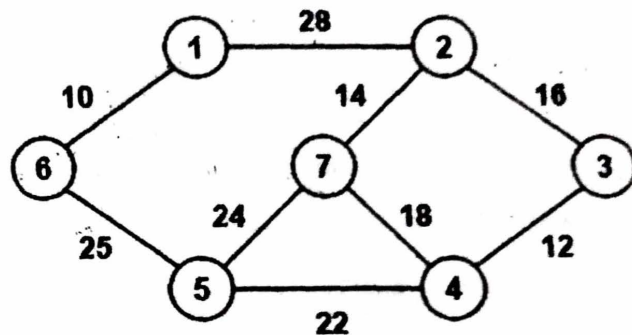
(c) Find an optimal solution to the knapsack instances 8

$$n = 7, m = 15$$

$$\text{Profit} = \{10, 5, 15, 7, 6, 18, 3\}$$

$$\text{Weight} = \{2, 3, 5, 7, 1, 4, 1\}$$

(d) Construct the minimum spanning tree (MST) for the given graph using Prim's and Kruskal's algorithm. 8



Unit-III

3. (a) What is dynamic programming concept? Give an example. 4

(b) Find an optimal paranthesization of a matrix chain product whose sequence of dimensions is $\langle 5, 10, 3, 12, 5, 50, 6 \rangle$. 8

(c) Explain Knuth Morris and Pratt (KMP) algorithm with suitable example. 8

(d) Explain Boyer Moore algorithm. Apply it to search for the pattern in the text given below : 8

[4]

T : 0 1 2 3 4 5 6 7 8 9

a b a c a a b a c c

P : 0 1 2 3 4 5

a b a c a b

Unit-IV

4. (a) What is backtracking and recursive backtracking? 4

(b) What is a Hamiltonian cycle? How does one apply backtracking algorithm to find Hamiltonian path and cycle? Explain. 8

(c) What is chromatic number? Give the state space tree for 4-coloring problem. Explain graph coloring problem. 8

(d) What is knapsack problem? Let us consider that the capacity of the knapsack $w = 60$ and the list of provided items are shown in the following table : 8

Item	A	B	C	D
Profit	280	100	120	120
Weight	40	10	20	24

What is the optimal solution and maximum profit?

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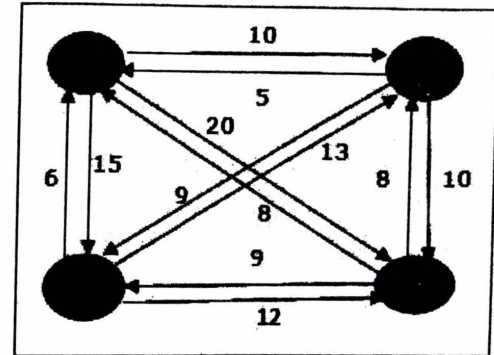
[5]

Unit-V

5. (a) Differentiate between P and NP problems. 4

(b) Explain the use of branch and bound method to solve 15 puzzle problem with least cost search. 8

(c) Define the traveling salesman problem. Illustrate the steps to solve the TSP given below : 8



(d) State and prove the Cook's theorem. 8

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**B. Tech. (Fourth Semester) Examination,
Nov.-Dec. 2023
(New Scheme)**

(Computer Science Engg. Branch)

DESIGN & ANALYSIS OF ALGORITHMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Marks : 35

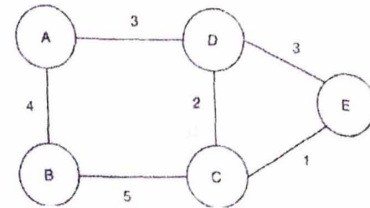
Note : Answer all questions. Part (a) of each question is compulsory and of 4 marks. Answer any two parts from part (b), (c) and (d), which is of 8 marks each.

1. (a) What do you mean by an Algorithm? Write the characteristics of the algorithm in brief.
- (b) What do you mean by order of magnitude? Explain

various types of asymptotic notations with suitable graphs.

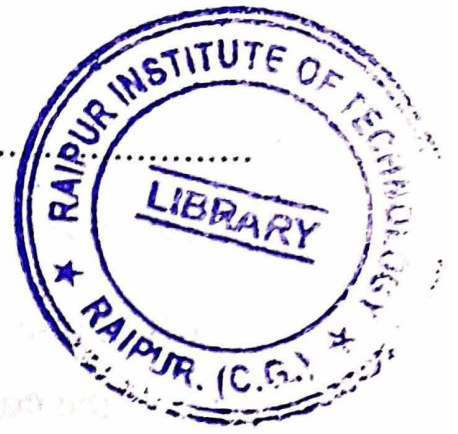
- (c) State the Master theorem in detail with example.
 - (d) What do you mean by sorting. Explain Heap Sort with an algorithm in detail.
2. (a) What do you mean by greedy approach? Describe in brief.
- (b) Write the detailed difference between quick sort and merge sort.
- (c) Find the optimal solution for the fractional knapsack problem by making use of the greedy approach. Consider :
- $$n = 5$$
- $$w = 60 \text{ kg}$$
- $$(w_1, w_2, w_3, w_4, w_5) = (5, 10, 15, 22, 25)$$
- $$(b_1, b_2, b_3, b_4, b_5) = (30, 40, 45, 77, 90)$$
- (d) What is the Spanning tree? Consider the following

graph and find the Minimum spanning tree.



3. (a) What do you mean by dynamic programming? Explain in brief.
- (b) Explain Boyer-Moore Algorithm in detail.
- (c) Explain Knuth Morris and Pratt algorithms in detail.
- (d) Write short notes on :
- (i) Longest common subsequence problems
 - (ii) Finite Automata for string matching problem
4. (a) Explain backtracking and recursive backtracking in brief.

- (b) Discuss the 8-queens problem with a suitable example.
 - (c) Explain the Hamiltonian cycle in detail with a suitable diagram.
 - (d) Explain graph coloring in detail.
5. (a) State Cook's theorem in brief.
- (b) What do you mean by NP-Hard problem? Explain at least one NP-hard problem in detail.
 - (c) Explain Least-Cost Search with example.
 - (d) Write the difference between NP-Hard and NP-Complete classes in detail.



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DESIGN & ANALYSIS OF ALGORITHM

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

***Note : Question (a) is compulsory and attempt any
two questions from (b), (c) & (d) in each
unit.***

Unit-I

1. (a) What do you mean by time complexity of an algorithm? 4
- (b) Explain Substitution method with example. 8
- (c) What is asymptotic notation? Explain its different

[2]

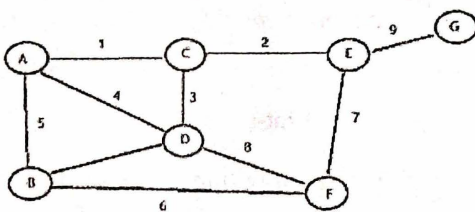
types. 8

- (d) Illustrate the operation of insertion sorting and define the complexity for worst case. 8

Unit-II

2. (a) What is Divide and Conquer strategy? 4
 (b) Explain merge sort algorithm and apply it to sort $\langle 1, 6, 3, 2, 7, 5, 8, 4 \rangle$. 8
 (c) What is minimum spanning tree?

Give the prims algorithm for finding minimum spanning tree for the given undirected graph. find the minimum spanning tree, their weight are attached with each edge. 8



- (d) Explain Knapsack problem with example. 8

Unit-III

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[3]

3. (a) Classify the steps used for development of dynamic programming. 4

- (b) Write down the algorithm for finding longest common subsequence and obtain the LCS for.

$X = \{B, A, C, D, B\}$, $Y = \{B, D, C, B\}$

- (c) Explain Boyer-Moore Algorithm with an example. 8

- (d) Write short notes on :

- (i) Matrix Chain multiplication.
 (ii) String matching algorithm.

Unit-IV

4. (a) What is the meaning of backtracking in field of algorithm? 4
 (b) Design a back tracking algorithm for the Hamiltonian Cycle. 8
 (c) Solve the 8-queen problem using backtracking method. 8
 (d) Write an algorithm for sum of subsets. Solve the

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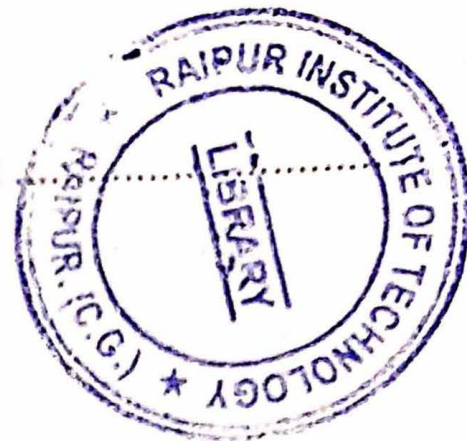
PTO

following problem $M = 30$ $W = \{5, 10, 13, 15, 18\}$.

8

Unit-V

5. (a) What is least-cost search? 4
- (b) Explain FIFO branch & bound. 8
- (c) Write short notes on : 8
 - (i) NP- completeness
 - (ii) Cook's theorem
- (d) Analyze NP-hard and NP-completeness with an example. 8



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Maximum Marks : 100

Minimum Marks : 35

Note : Answer all questions. Part (a) of each question is compulsory and of 4 marks. Answer any two parts from part (b), (c) and (d), which is of 8 marks each.

1. (a) What do you mean by Asymptotic Notations? Also write types of notations.

4

(b) Consider the following functions and solve them using Master's theorem

8

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[2]

(i) $T(n) = 4T(n/2) + n$

(ii) $T(n) = 2T(n/2) + n$

- (c) Illustrate the following array elements and re-arrange them in ascending order using Insertion Sort.

16, 15, 4, 13, 2, 1

8

- (d) Consider the following array elements and find the location of key element from the array, where key element = 20.

10, 50, 30, 70, 80, 60, 20, 90, 40

using binary search algorithm.

8

2. (a) Write difference between Kruskal & Prim's algorithm. 4

- (b) Sort the following element using Merge Sort algorithm.

8

20, 30, 50, 10, 5, 60, 80, 8, 64, 75, 32, 69

- (c) Sort the following element using Quick Sort algorithm.

3, 4, 7, 1, 5, 9, 2, 8, 4

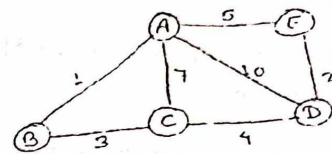
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[3]

- (d) Solve the following by Kruskal algorithm.

8



3. (a) Discuss basic divide & conquer for matrix multiplication. 4

- (b) Discuss KMP algorithm and compute the following :

$T : a b a b c a b c a b a b a b d$

$P : a b a b d$

8

- (c) For the given sequence {4, 10, 3, 12, 20 and 7}.

The matrices have size 4×10 , 10×3 , 3×12 , 12×20 , 20×7 . Find out optimal matrix chain multiplication sequence dynamic programming. 8

- (d) Discuss Boyer Moore pattern searching algorithm and compute the following :

T : This is a test

P : test

8

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PTO

[2]

(i) $T(n) = 4T(n/2) + n$

(ii) $T(n) = 2T(n/2) + n$

- (c) Illustrate the following array elements and re-arrange them in ascending order using Insertion Sort.

16, 15, 4, 13, 2, 1

8

- (d) Consider the following array elements and find the location of key element from the array, where key element = 20.

10, 50, 30, 70, 80, 60, 20, 90, 40

using binary search algorithm.

8

2. (a) Write difference between Kruskal & Prims algorithm. 4

- (b) Sort the following element using Merge Sort algorithm.

8

20, 30, 50, 10, 5, 60, 80, 8, 64, 75, 32, 69

- (c) Sort the following element using Quick Sort algorithm.

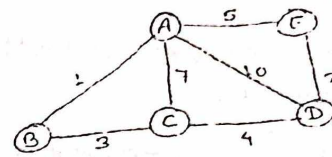
3, 4, 7, 1, 5, 9, 2, 8, 4

8

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[3]

- (d) Solve the following by Kruskal algorithm. 8



3. (a) Discuss basic divide & conquer for matrix multiplication. 4

- (b) Discuss KMP algorithm and compute the following :

$T : a b a b c a b c a b a b a b d$

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$T : \text{This is a test}$

$P : \text{test}$

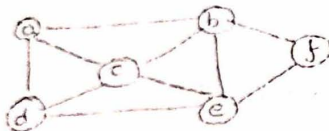
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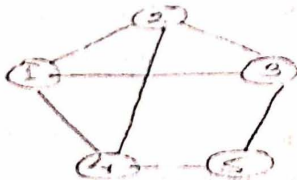
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[4]

4. (a) What do you understand by Backtracking in algorithm? 4
- (b) Consider the following set $S: \{1, 3, 4, 5\}$ and sum = 8. Find the subset(s). 8
- (c) Form a Hamiltonian cycle/circuit from the given undirected graph. 8



- (d) Color the following undirected graph, using Graph Coloring concept where chromatic no. is 3. 8



5. (a) Define FIFO and LC (Least-Cost) search. 4

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[5]

- (b) Write short notes on the following : 8
- (i) P-Class
 - (ii) NP-Class
 - (iii) NP-Hard
 - (iv) NP-Complete
- (c) Explain Cook's theorem in details. 8
- (d) Consider the following values/distances of the city, find the optimal solution for the travelling salesman problem. 8

	A	B	C	D
A	∞	10	5	3
B	8	∞	9	7
C	1	6	∞	9
D	2	3	8	∞

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**B. Tech. (Fourth Semester) Examination,
April-May 2025
(New Scheme)**

(Computer Science Engg. Branch)

DESIGN & ANALYSIS of ALGORITHMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Marks : 35

***Note : Attempt all questions. Part (a) is compulsory
from each unit. Attempt any two parts from
(b), (c) and (d).***

Unit-I

1. (a) What is Big O notation? 4
- (b) State the master theorem and explain it with examples. 8

- (c) Explain heap sort algorithm with example. 8
- (d) Solve the following recursion equation by using master method : 8
- (i) $T(n) = 3T(n/2) + n^2$
- (ii) $T(n) = 2T(n/2) + n \log n$

Unit-II

2. (a) Differentiate between divide and conquer algorithm and Greedy method. 4
- (b) Explain & analyse merge sort algorithm. 8
- (c) Explain Huffmann algorithm with suitable example. 8
- (d) Explain in short Kruskal & prims algorithm with example. 8

Unit-III

3. (a) What is dynamic programming? 4
- (b) Explain Boyer-Moore algorithm with example. 8

[3]

- (c) Write short note on Knuth Morris and Pratt algorithms. 8
- (d) Explain the following by using Matrix chain multiplication. $\{4, 10, 3, 12, 20, 7\}$. 8

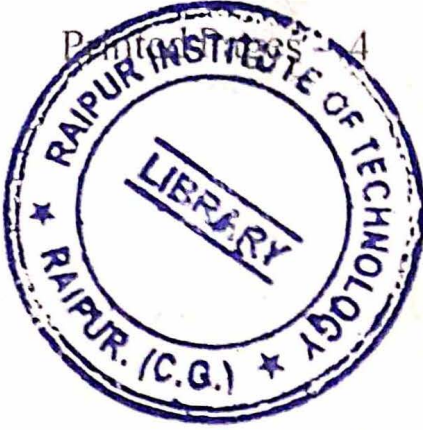
Unit-IV

4. (a) What is backtracking? Write any two application of backtracking. 4
- (b) Design a back tracking algorithm for Hamiltonia cycle problem. 8
- (c) Find all the possible solution for 8×8 chess board, 8 Queen problem. 8
- (d) Write short notes on : 8
- (i) Sum of subset
 - (ii) Graph coloring

Unit-V

5. (a) What do you mean by LC (Least Cost) search? 4

- (b) Explain TSP with example. 8
- (c) Write a short note on following : 8
- (i) NP-completeness
 - (ii) Cook's theorem
- (d) Explain 15 puzzle problem with suitable example. 8



Roll No. :



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**B. Tech. (Fourth Semester) Examination,
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(Computer Science Engg. Branch)

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Time Allowed : Three hours

Maximum Marks : 100

Minimum Marks : 35

*Note : Attempt all questions. Part (a) is compulsory
from each unit carry 4 marks. Attempt any
two parts from (b), (c) and (d) carry 8 marks.*

Unit-I

1. (a) What is Algorithm? Write its criteria and distinct area? 4
- (b) What are asymptotic notations? Explain different types of Asymptotic notations. 8

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PTO

[2]

- (c) Solve recurrence using recursion tree method. 8

$$T(n) = 2T(n/2) + n^2$$

- (d) Solve the following Recurrence by Master Method : 8

(i) $T(n) = 4T(n/2) + n^3$

(ii) $T(n) = 32T(n/2) + n^3$

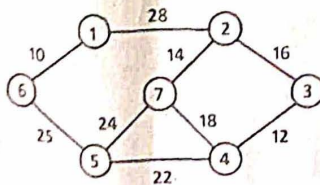
(iii) $T(n) = 9T(n/3) + n$

(iv) $T(n) = 7T(n/3) + n^2$

Unit-II

2. (a) Explain divide and conquer approach for solving problem. 4

- (b) Find the minimal spanning Tree Using Prim's algorithm for following graph : 8



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[3]

- (c) Consider 5 items along their respective weights and values : 8

$$I = \langle I_1, I_2, I_3, I_4, I_5 \rangle$$

$$W = \langle 5, 10, 20, 30, 40 \rangle$$

$$V = \langle 30, 20, 100, 90, 160 \rangle$$

The capacity of Knapsack $W = 60$. Find the solution to the fractional; Knapsack problem.

- (d) Sort the following elements using radix sort. 8

170, 45, 75, 90, 802, 24, 2, 66

Unit-III

3. (a) Explain Dynamic Programming. 4

- (b) Explain Boyer-Moore algorithm with its complexity. 8

- (c) Explain Knuth Morris and Pratt algorithms. 8

- (d) Using Dynamic Programming to find the LCS two sequences are : 8

X : A C A D B

Y : C B D A

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PTO

Unit-IV

4. (a) Explain Back tracking. 4
- (b) Explain 8-queens problem and solve it using backtrack technique. 8
- (c) Explain Graph Colorings method with example. 8
- (d) Given a set $S = \{5, 10, 12, 13, 15, 18\}$ and $d = 30$ obtain the subset sum using backtrack approach. 8

Unit-V

5. (a) Explain NP-Hard and NP complete problems. 4
- (b) Explain travelling sales person problem. 8
- (c) Explain FIFO branch and bound technique. 8
- (d) Explain Cook's theorem. 8



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Roll No. :

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Time Allowed : Three hours

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Note : Answer all questions. Part (a) of each question is compulsory and of 4 marks. Answer any two parts from part "b", "c" and "d", which is of 8 marks each.

1. (a) What are the fundamental steps involved in problem solving using algorithm.
- (b) Explain the Asymptotic notations used in designing algorithms.

[2]

(c) Write the algorithm of Insertion Sort and analyze this.

(d) Solve the following recurrences.

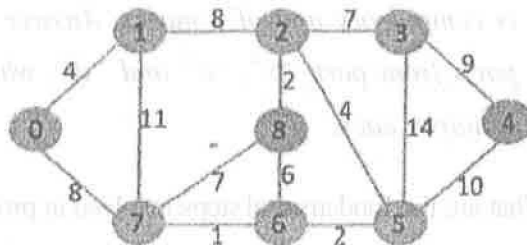
(i) $T(n) = T(n/2) + T(n/4) + T(n/8) + n$

(ii) $T(n) = T(n-2) + 2 \lg n$

2. (a) Write the general plan for divide-and-conquer algorithms. Write any four example of divide-and-conquer.

(b) Write the algorithm of Quick Sort and analyze it.

(c) Find the minimum spanning tree using Kruskal algorithm for the following graph.



(d) Solve the Knapsack problem using Greedy algorithm, for the following data.

[3]

Item	A	B	C	D
Profit	280	100	120	120
Weight	40	10	20	24

where $W = 60$.

3. (a) What is Dynamic programming? Write the steps of Dynamic Programming.

(b) Find Longest Common subsequence of X and Y, where $X = A, B, C, B, D, A, B$ and $Y = B, D, C, A, B, A$.

(c) Find the pattern in the given text by using Boyer Moore algorithm, where Text : "G C A A T G C C T A T G T G A C C" and Pattern : 'T A T G T G'.

(d) Write any one application of string matching algorithm. Which string matching algorithm you will use for this? How you will think that your selected algorithm is better than other string matching algorithm.

4. (a) Explain Backtracking and recursive backtracking.

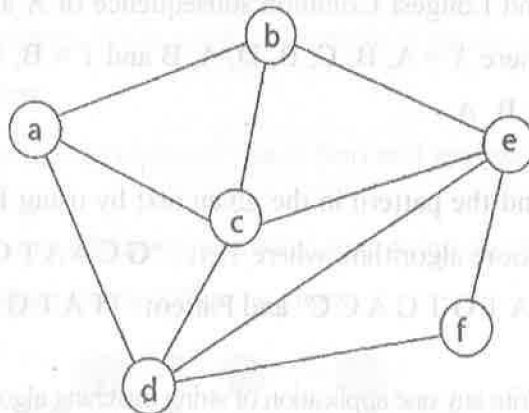
(b) Write the rules of 8-queen problem. Find any one

[4]

solution of 8-queen problem by using backtracking.

- (c) Find the solution for the sum of subsets problem with the set $S = \{3, 4, 5, 6, 10\}$ with $M = 9$ using backtracking method.

- (d) Find the Hamiltonian cycle for the given graph using backtracking.



5. (a) Define Branch-and-Bound method with different types of nodes used.

- (b) Solve the 0/1 knapsack problem using Branch and Bound with following data :

[5]

Items	Weight	Value
11	3	4
12	4	5
13	5	6

where $W = 7$.

- (c) Explain FIFO Branch and Bound and LC Branch and Bound, with example.

- (d) Write short notes on any **two** :

- (i) Cook's theorem
- (ii) NP Complete Classes
- (iii) NP Hard Classes

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Roll No. :

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1. (a) What are the fundamental steps involved in problem solving using algorithm.
- (b) Explain the Asymptotic notations used in designing algorithms.

[2]

(c) Write the algorithm of Insertion Sort and analyze this.

(d) Solve the following recurrences.

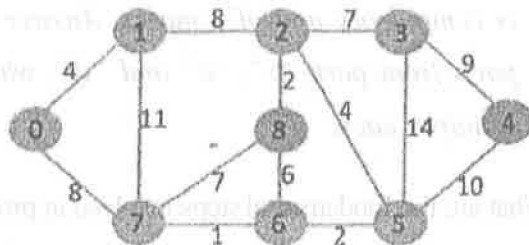
(i) $T(n) = T(n/2) + T(n/4) + T(n/8) + n$

(ii) $T(n) = T(n-2) + 2 \lg n$

2. (a) Write the general plan for divide-and-conquer algorithms. Write any four example of divide-and-conquer.

(b) Write the algorithm of Quick Sort and analyze it.

(c) Find the minimum spanning tree using Kruskal algorithm for the following graph.



(d) Solve the Knapsack problem using Greedy algorithm, for the following data.

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[3]

Item	A	B	C	D
Profit	280	100	120	120
Weight	40	10	20	24

where $W = 60$.

3. (a) What is Dynamic programming? Write the steps of Dynamic Programming.

(b) Find Longest Common subsequence of X and Y, where $X = A, B, C, B, D, A, B$ and $Y = B, D, C, A, B, A$.

(c) Find the pattern in the given text by using Boyer Moore algorithm, where Text : "G C A A T G C C T A T G T G A C C" and Pattern : 'T A T G T G'.

(d) Write any one application of string matching algorithm. Which string matching algorithm you will use for this? How you will think that your selected algorithm is better than other string matching algorithm.

4. (a) Explain Backtracking and recursive backtracking.

(b) Write the rules of 8-queen problem. Find any one

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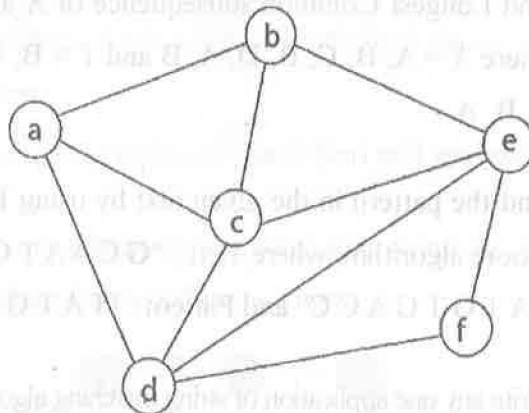
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[4]

solution of 8-queen problem by using backtracking.

- (c) Find the solution for the sum of subsets problem with the set $S = \{3, 4, 5, 6, 10\}$ with $M = 9$ using backtracking method.

- (d) Find the Hamiltonian cycle for the given graph using backtracking.



5. (a) Define Branch-and-Bound method with different types of nodes used.

- (b) Solve the 0/1 knapsack problem using Branch and Bound with following data :

[5]

Items	Weight	Value
11	3	4
12	4	5
13	5	6

where $W = 7$.

- (c) Explain FIFO Branch and Bound and LC Branch and Bound, with example.

- (d) Write short notes on any **two** :

- (i) Cook's theorem
- (ii) NP Complete Classes
- (iii) NP Hard Classes

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**B. Tech. (Fourth Semester) Examination,
April-May 2022
(AICTE Scheme)**

(Computer Science Engg. Branch)

DESIGN & ANALYSIS OF ALGORITHMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Marks : 35

*Note : Attempt all questions. Attempt any two parts
from (a), (b) and (c) of each unit. Each part
carries 10 marks.*

Unit-I

1. (a) Explain asymptotic notations used in designing algorithm.
- (b) Write the algorithm of Bubble sort and analyze it.

[2]

(c) Explain master method and solve the following using master slave method :

(i) $T(n) = 3T(n/2) + n^2$

(ii) $T(n) = 2T(n/2) + n \log n$

(iii) $T(n) = \sqrt{2} T(n/2) + \log n$

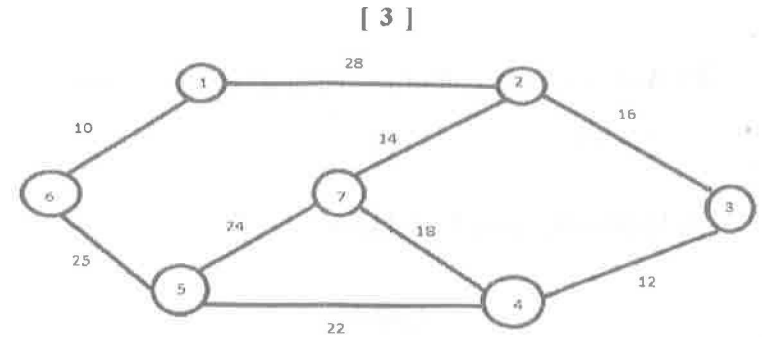
Unit-II

2. (a) Write the algorithm of Merge sort and analyze it.

(b) Solve the following using divide and conquer algorithm for matrix multiplication :

$$A = \begin{bmatrix} 1 & 3 \\ 7 & 5 \end{bmatrix}, B = \begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix}$$

(c) Find the minimal spanning tree using Prim's algorithm for following graph :



Unit-III

3. (a) Given two sequences X [1 m] and Y [1.....n].

Find the longest common sequence to both :

X	A	B	C	B	D	A	B
Y	B	D	C	A	B	A	

(b) Give the matrix having {4, 10, 3, 12, 20 and 7}, size 4×10 , 10×3 , 3×12 , 12×20 , 20×7 so compute the optimal solution using Matrix Chain multiplication.

(c) Explain Boyer Moore algorithm with its complexity. Also shows the complexity comparison of string matching algorithm.

Unit-IV

4. (a) Explain Hamilton cycle with an example.

[4]

(b) What is Graph colouring method? Explain with an example.

(c) Explain Knapsack problem.

Unit-V

5. (a) For the given set of items and knapsack capacity = 5 kg, find the optimal solution for the 0/1 knapsack problem making use of dynamic programming approach :

Item	Weight	Value
1	2	3
2	3	4
3	4	5
4	5	6

(b) Explain travelling sales person problem with an example.

(c) Write short note on :

(i) NP Complete Classes

(ii) FIFO Branch and Bound